

Outline

The overtuning problem

Model selection — across learners, HP configs, feature sets, preprocessings — is core to ML.

Trap: if we evaluate our *final* selected learner on the same resampling splits that we used to *select* it, information about the test data leaks into the model-building process.

- ▶ Result: the reported CV score is too optimistic
- ▶ Worse with many HP configurations / large search spaces
- ▶ Worse on small datasets where CV scores are already noisy
- ▶ The more we tune, the more we “cheat”

The untouched test set principle

To get an honest estimate of generalization error, the test set must be:

- ▶ Not used for training
- ▶ Not used for HP selection
- ▶ Not used for feature selection
- ▶ Not used for any decision about the model

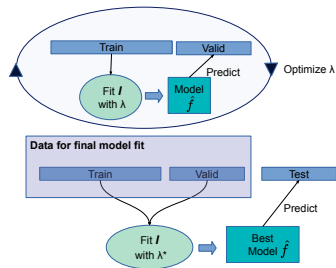
In short: the test set is *looked at exactly once*, at the very end.

Two practical implementations:

1. **Train / validation / test split** — 3-way data split
2. **Nested resampling** — 3-way idea generalized via two nested CV loops

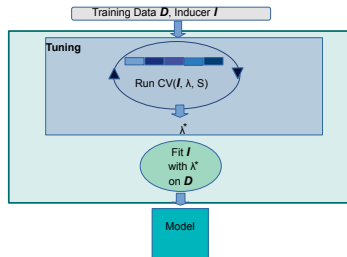
The 3-way split

- ▶ Split data into **train**, **validation**, and **test**
- ▶ Tune by training on *train*, scoring on *validation*
- ▶ Pick best HP λ^*
- ▶ Refit on train+validation
- ▶ Score *once* on *test* → unbiased GE estimate



Tuning becomes part of training

- ▶ Conceptually: the train+validation procedure is the “real” training algorithm
- ▶ Hyperparameters move from being *inputs* of training to being *outputs* (the algorithm now “self-tunes”)
- ▶ This view makes nesting natural: just resample the outer level



Limitation of 3-way split

- ▶ Only *one* validation set $\Rightarrow$ noisy HP selection
- ▶ Only *one* test set $\Rightarrow$ noisy GE estimate
- ▶ Both worse on small datasets

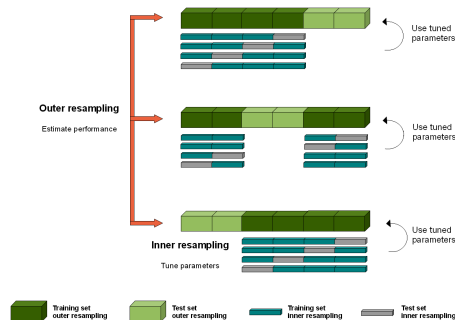
Solution: replace each split with a *resampling loop*. That gives **nested resampling**.

Two nested resampling loops

- **Outer loop:** estimate GE
- **Inner loop:** tune HPs (returns the best λ^* for the outer training split)

Example: 3-fold outer CV with 4-fold inner CV.

- In each outer iteration, the dark-green part is the training set for “self-tuning”
- Inside it, 4-fold CV picks λ^*
- Then refit on the full dark-green part, score once on light-green



Per-iteration mechanics

In each outer iteration:

1. Split off the outer test fold (light green)
2. On the outer training part (dark green):
 - ▶ Run inner CV across all candidate λ_i
 - ▶ Return the winner λ^* (best mean inner-CV score)
3. Refit the model with λ^* on the *full* outer training part
4. Evaluate *once* on the outer test fold

Aggregate the outer test-fold scores $\rightarrow$ honest estimate of GE for the tuned learner.

Why nested resampling is unbiased

The outer test folds were:

- ▶ Never used to fit model parameters
- ▶ Never used to select hyperparameters
- ▶ Never seen during inner CV

⇒ The outer-fold scores are honest estimates of how the tuned procedure would perform on truly fresh data.

Cost

Nested resampling is expensive:

- ▶ For each of k_{outer} outer folds: train $|\Lambda| \cdot k_{\text{inner}}$ models
- ▶ Plus one final refit per outer fold
- ▶ Total: $k_{\text{outer}} \cdot (|\Lambda| \cdot k_{\text{inner}} + 1)$ model fits

Practical defaults:

- ▶ Outer: 3- or 5-fold CV (sometimes just hold-out if data is large)
- ▶ Inner: 5-fold CV, or random search with a fixed budget
- ▶ Random search inside the inner loop scales better than grid for many HPs

When you really need nested resampling

- ▶ You will report a single performance number for “your method”
- ▶ You’ve tuned multiple HPs, possibly searched over many models
- ▶ Dataset is small enough that a single hold-out is noisy
- ▶ You’re comparing your method against alternatives — the comparison must be honest

When you can skip it:

- ▶ Very large data: a single hold-out test set is fine
- ▶ Final deployed model: you already chose HPs; just refit on all data
- ▶ Exploratory work: be honest about “this is a CV score after tuning, not a true GE estimate”

Common mistakes

- ▶ Tuning then reporting the best inner-CV score as “test error”. **Optimistic bias** that grows with search size.
- ▶ Doing feature selection on the full dataset *before* CV. **Leakage**.
- ▶ Standardizing / imputing / oversampling using the full dataset. **Leakage**.
- ▶ Choosing a metric *after* seeing CV results. **Selection bias**.

Rule: *anything* that uses the labels must be inside the CV loop. If unsure, ask “would this be the same in production where I don’t have the labels yet?”

Wrap-up

1. Tuning + evaluation on the same data $\rightarrow$ optimistic bias
2. Untouched test set principle: test data is looked at exactly once
3. 3-way split (train/valid/test) is the simplest implementation
4. Nested resampling generalizes this with inner+outer CV loops
5. Expensive but the only honest way to report a tuned learner's GE on small data